

Unit 11, Lesson 1: Exploring Exponential Growth Versus Decay



Benchmark

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.



Learning Targets

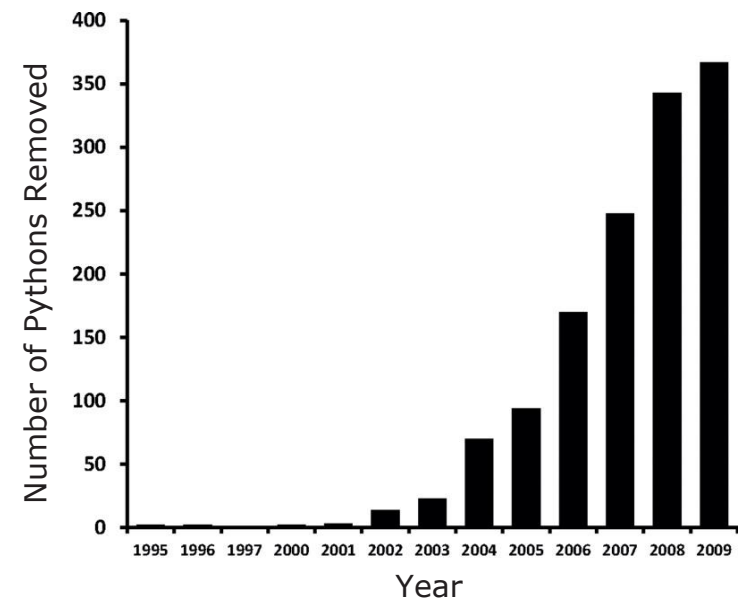
- ☐ I can classify a function as representing exponential growth or decay from a graph.
- ☐ I can classify a function as representing exponential growth or decay from a table.



11.1.1 Warm-Up: Invasive Snakes

1. The Burmese python is one of the largest species of snakes and is one of the most concerning invasive species in Everglades National Park. The graph below shows the number of pythons removed each year from 1995 to 2009.

Scientists now estimate the population of Burmese pythons in the Everglades is more than 100,000.



- a. What do you notice about the graph?
- b. Given the graph, are you surprised scientists now estimate more than 100,000 snakes in Everglades National Park?



11.1.2 Exploration: Dominoes and Stickers

1. Morris, a physics professor at the University of Toronto, set up a series of 13 dominoes, each roughly $1\frac{1}{2}$ times the size of the one knocking it over. The original domino is 5 millimeters (mm) tall and only 1 mm thick. The 13th domino is 3.2 feet (ft) tall and weighs about 100 pounds.



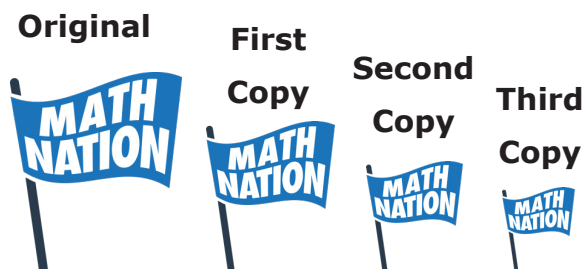
- a. What do you wonder about the dominoes?
- b. Do you think starting with the first domino, which is smaller than a Tic Tac, will result in the 13th domino falling?

2. In a 1983 edition of the American Journal of Physics, Lorne Whitehead wrote about his discovery that as a domino falls it could topple things larger than itself. He found that a domino is capable of knocking down a domino that is 50% larger than itself.
 - a. Use the information about the 13th domino to determine the height, in feet (ft), for the 14th and 15th dominos.

Domino	Height (ft)
13 th	3.2
14 th	
15 th	

- b. Discuss with your partner how the description that Dr. Morris uses, " $1\frac{1}{2}$ times itself," and the description Dr. Whitehead uses, "50% larger than itself," mean the same thing. Write your explanation or give an example to explain.

3. Ashley used a copy machine to make copies of a Math Nation sticker to observe how the size changes. Before making copies, she pressed the 75% zoom button.



- She made the first copy of the sticker.
- To make the second copy, she removed the original sticker from the copy machine and replaced it with the first copy of the sticker. She then pressed copy.
- To make the third copy, she removed the first copy of the sticker and replaced it with the second copy of the sticker. She then pressed copy.

How is each copy of the sticker related to the copy that was made before?



11.1.3 Collaboration: Graphs and Tables

There are two types of exponential functions: exponential growth and exponential decay.



An **exponential function** is a function with a constant percent rate of change.



Exponential growth describes an exponential function whose constant percent rate of change is greater than 100%.



Exponential decay describes an exponential function whose constant percent rate of change is greater than 0% but less than 100%.

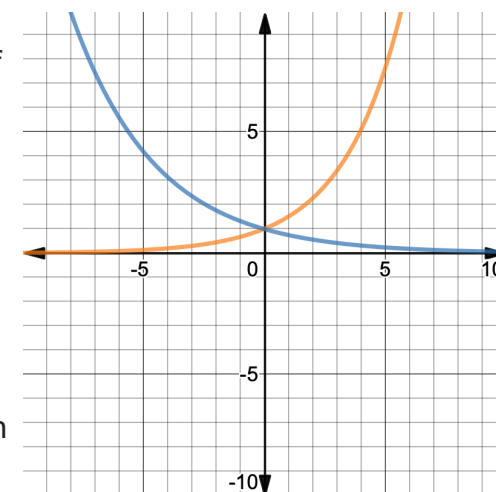
1. Use the descriptions of exponential growth and decay to determine what the orange graph represents and what the blue graph represents.

Orange:

- Exponential growth
- Exponential decay

Blue:

- Exponential growth
- Exponential decay



2. One of the graphs models the size of the dominoes, and the other models the size of the sticker. Discuss with your partner how to determine which graph, orange or blue, is the function that models the size of the dominoes. Write a summary of your discussion.

3. Each graph assumes the first sticker or the first domino is considered 100% of the original size.

What does the point $(0, 1)$ represent for the function that represents the dominoes?

4. The tables represent the two different contexts. Each table assumes that the first sticker or the first domino is 100% of the original size.

Table 1	
x	y
0	1
1	$\frac{3}{2}$
2	$\frac{9}{4}$
3	$\frac{27}{8}$
4	$\frac{81}{16}$
5	

Table 2	
x	y
0	1
1	$\frac{3}{4}$
2	$\frac{9}{16}$
3	$\frac{27}{64}$
4	$\frac{81}{256}$
5	

- a. In Table 1, find the y -value for $x = 5$ using the y -value for $x = 4$. Then, explain how you determined your answer.
- b. Elise found that in Table 1, when she divided $\frac{9}{4}$ by $\frac{3}{2}$ or $\frac{27}{8}$ by $\frac{9}{4}$, she got the same value. How does the value she found relate to the context?

- c. Use Elise's method and divide consecutive y -values in Table 2. How does the value relate to the context?

5. Discuss with your partner which table matches each context. Then, complete the statements based on your discussion.

Table 1 matches the ☐ stickers ☐ dominoes context because as

the x -values increase, the y -values in Table 1 are

☐ increasing ☐ decreasing at a constant percent rate

☐ between 0% and 100%.
☐ greater than 100%.

Table 2 matches the ☐ stickers ☐ dominoes context because as

the x -values increase, the y -values in Table 2 are

☐ increasing ☐ decreasing at a constant percent rate

☐ between 0% and 100%.
☐ greater than 100%.



11.1.4 Bring It Together: Exponential Equations



An exponential equation is a type of equation in which the input is an exponent. It can be written in the form $y = a(b)^x$.

The a -value of an exponential equation is the initial amount.

From the previous contexts with the first sticker and first domino, those were considered 100% or 1. However, this is not the case for every situation. For example, the **dimensions** of the sticker or the **height** of the first domino could have been used as the initial value.

The b -value of an exponential equation is the growth or decay factor.

Exponential Growth	Exponential Decay
b is the growth factor	b is the decay factor
$b = 1 + r$, where r is the growth rate	$b = 1 - r$, where r is the decay rate
$y = a(1 + r)^x$	$y = a(1 - r)^x$

1. Complete the statement.

When the equation is written in the form $y = a(b)^x$, the

b is greater than 1 when the equation is modeling

exponential ☐ decay ☐ growth, and is between 0 and 1 when

the equation is modeling exponential ☐ decay. ☐ growth.

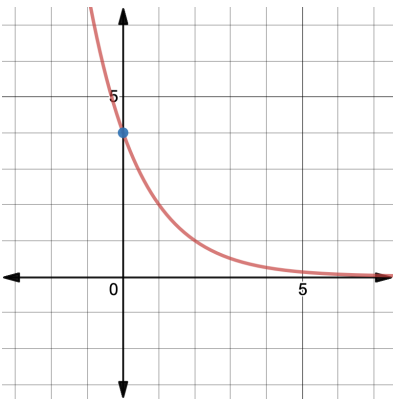
2. The equation that models the dominoes situation is $y = 1\left(1 + \frac{1}{2}\right)^x$ or $y = \left(\frac{3}{2}\right)^x$. The equation that models the sticker situation is $y = 1\left(1 - \frac{1}{4}\right)^x$ or $y = \left(\frac{3}{4}\right)^x$. Discuss with your partner how the value of b in each equation relates to the relationship observed in the tables.



11.1.5 Your Turn!

1. Determine if each representation of a function is an example of exponential decay or exponential growth. Justify your selection.

a.

Representation	Type of Exponential Function
	☐ Exponential decay ☐ Exponential growth
Justification	

b.

Representation	Type of Exponential Function
$y = 50(1 + 0.85)^x$	☐ Exponential decay ☐ Exponential growth
Justification	

c.

Representation	Type of Exponential Function												
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>-3</td><td>4</td></tr> <tr><td>-2</td><td>8</td></tr> <tr><td>-1</td><td>16</td></tr> <tr><td>0</td><td>32</td></tr> <tr><td>1</td><td>64</td></tr> </tbody> </table>	x	y	-3	4	-2	8	-1	16	0	32	1	64	○ Exponential decay ○ Exponential growth
x	y												
-3	4												
-2	8												
-1	16												
0	32												
1	64												
Justification													

d.

Representation	Type of Exponential Function
$y = 50(0.85)^x$	○ Exponential decay ○ Exponential growth
Justification	

e.

Representation	Type of Exponential Function												
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>0</td><td>$\frac{9}{2}$</td></tr> <tr><td>1</td><td>$\frac{9}{4}$</td></tr> <tr><td>2</td><td>$\frac{9}{8}$</td></tr> <tr><td>3</td><td>$\frac{9}{16}$</td></tr> <tr><td>4</td><td>$\frac{9}{32}$</td></tr> </tbody> </table>	x	y	0	$\frac{9}{2}$	1	$\frac{9}{4}$	2	$\frac{9}{8}$	3	$\frac{9}{16}$	4	$\frac{9}{32}$	○ Exponential decay ○ Exponential growth
x	y												
0	$\frac{9}{2}$												
1	$\frac{9}{4}$												
2	$\frac{9}{8}$												
3	$\frac{9}{16}$												
4	$\frac{9}{32}$												
Justification													



11.1.6 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...

Unit 11, Lesson 2: Key Features of Exponential Functions



Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



Learning Targets

- ☐ I can determine the key features of an exponential function from a table, graph, or exponential function.
- ☐ I can classify a function as representing exponential growth or decay.